ALTERNATING CURRENT

CHAPTER SEVEN
CLASS - XII

MANISH JOSHI

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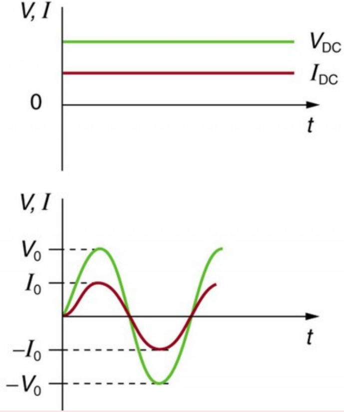
October 22, 2024

Learning Objectives:

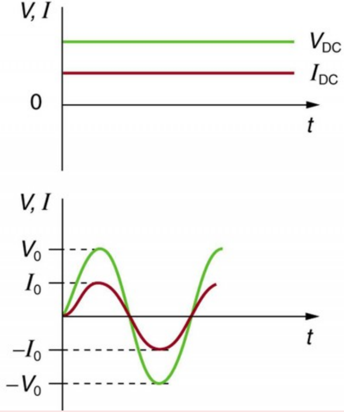
- Alternating currents,
- Peak and RMS value of alternating current/voltage;
- Reactance and impedance;
- LCR series circuit (phasors only),
- Resonance,
- Power in AC circuits, power factor, wattless current.
- Transformer.

Types of Electric Current



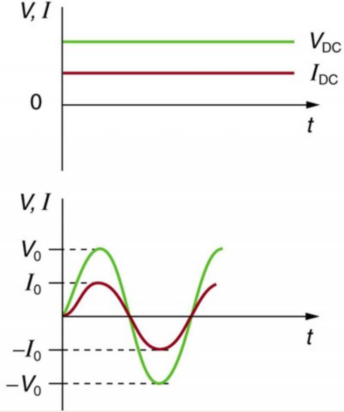
(1) Alternating Current (AC) :

Types of Electric Current



(1) Alternating Current (AC) : An alternating current is one which **periodically changes its magnitude and direction**.

Types of Electric Current



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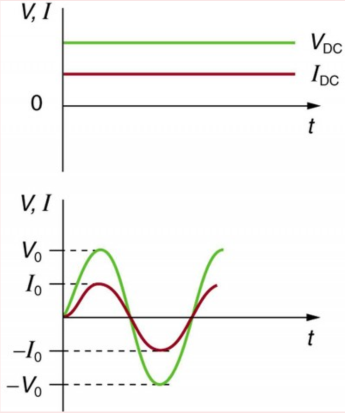
The electric mains supply in our homes and offices is a **voltage** that **varies like a sine function with time**.
Such a voltage is called **alternating voltage** (ac voltage) and the current driven by it in a circuit is called the **alternating current** (ac current).

Types of Electric Current



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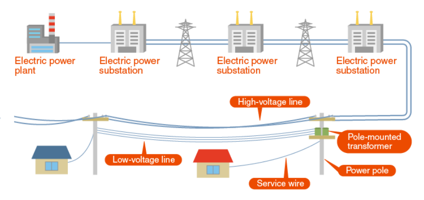
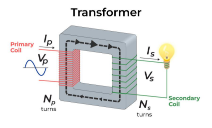
The electric mains supply in our homes and offices is a **voltage** that **varies like a sine function with time**. Such a voltage is called **alternating voltage** (ac voltage) and the current driven by it in a circuit is called the **alternating current** (ac current).

(2) Direct Current (DC) : Direct current which may or may not change in magnitude but **does not change its direction**.

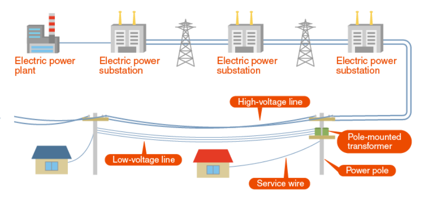
Advantages of AC over DC

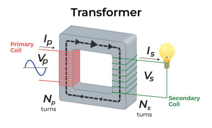




- Alternating voltage **can be transmitted to distant places** with **very little loss** of **electric energy**.

Advantages of AC over DC





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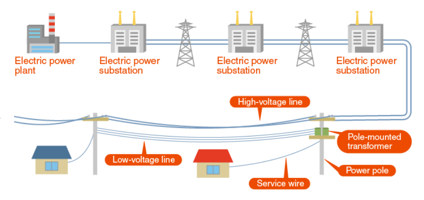
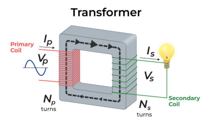
Advantages of AC over DC





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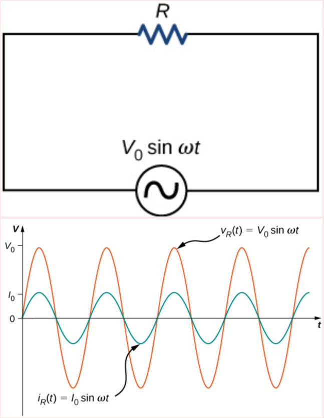
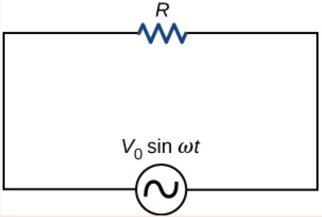
- AC voltages can be easily and **efficiently converted from one voltage to the other** by means of **transformers**.
- **AC** can be **easily converted into DC** by **using rectifiers**.

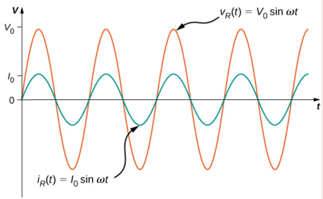
AC VOLTAGE APPLIED TO A RESISTOR



- First, consider a **resistor connected across an ac voltage**.

AC VOLTAGE APPLIED TO A RESISTOR





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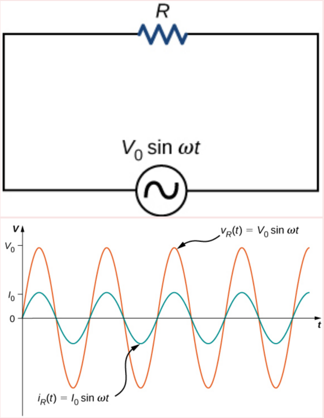

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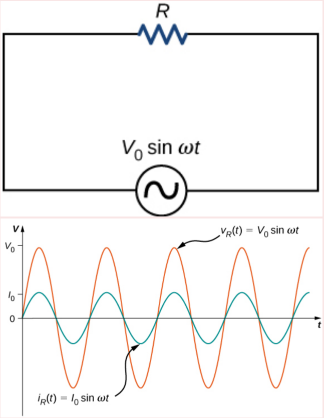

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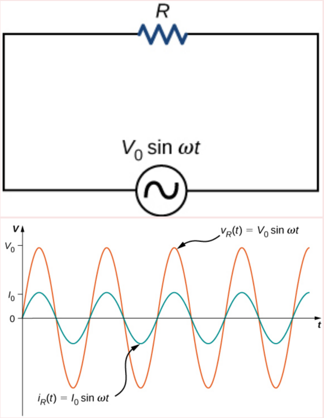

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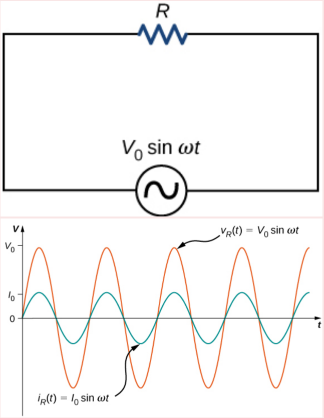

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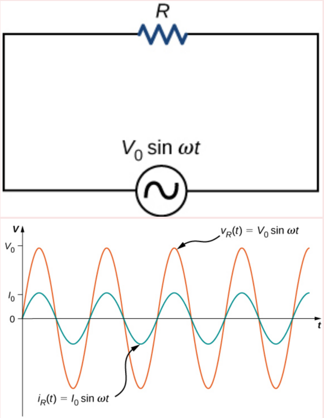

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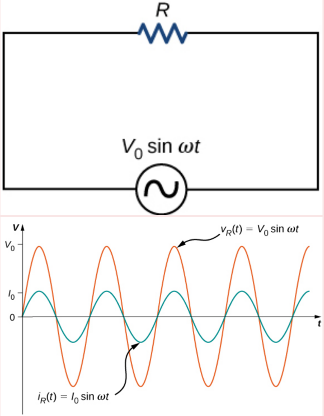
$$i_R(t) = \frac{v_R(t)}{R} = \frac{V_0}{R} \sin \omega t = I_0 \sin \omega t$$

AC VOLTAGE APPLIED TO A RESISTOR



- Here, $I_0 = \dfrac{V_0}{R}$ is the **amplitude of** the time-varying **current**.
- Plots of $i_R(t)$ and $v_R(t)$ are shown in Figure. **Both curves reach their maxima and minima at the same times,** that is, the **current through and the voltage across the resistor are in phase**.

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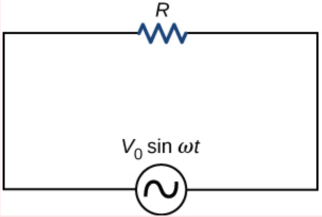

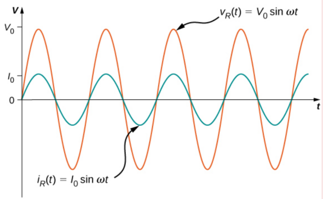

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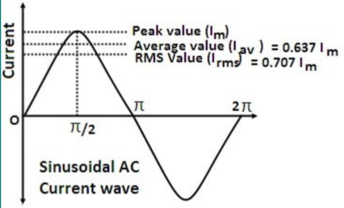
Some definitions regarding AC



Peak value of Alternating Current/Voltage:

The **peak value of an AC waveform refers to the maximum positive or negative value reached during each cycle**. It represents the **highest magnitude of the waveform**. Generally, this is represented as $\boldsymbol{I_m}$ or $\boldsymbol{I_0}$.

Average/Mean value of Alternating Current:

AC doesn't have a constant value, as it oscillates. Thus, **we use the average value of AC over a half cycle**, as **for the complete cycle average value is 0** because **positive and negative halves cancel each other out**.

Some definitions regarding AC

Average/Mean value of Alternating Current:

For positive half cycle.

$$I_{av} = \frac{1}{T/2}\int_0^{T/2} I\,dt$$

Putting $I = I_0 \sin \omega t$ and $T = \dfrac{2\pi}{\omega}$

$$I_{av} = \frac{1}{\pi/\omega}\int_0^{\pi/\omega} I_0 \sin \omega t\,dt = \frac{\omega}{\pi}\frac{I_0}{\omega}\left[-\cos \omega t\right]_0^{\pi/\omega} = -\frac{I_0}{\pi}(\cos \pi - \cos 0)$$

$$\boxed{I_{av} = -\frac{I_0}{\pi}(-1-1) = \frac{2}{\pi}I_0}$$

Some definitions regarding AC

Root Mean Square (RMS) Value of Alternating Current:

RMS value is **the value of DC which is flowing through the conductor to produce the same amount of heat as AC is flowing through the same conductor**.

Example 1

A light bulb is rated at 100 W for a 220 V supply. Find (a) the resistance of the bulb; (b) the peak voltage of the source; and (c) the rms current through the bulb.

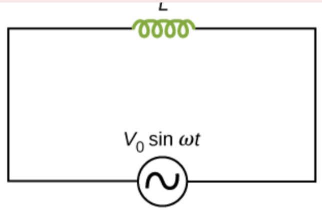
Exercise 1

A 100 Ω resistor is connected to a 220 V, 50 Hz ac supply.
(a) What is the rms value of current in the circuit?
(b) What is the net power consumed over a full cycle?

Exercise 2

(a) The peak voltage of an ac supply is 300 V. What is the rms voltage?
(b) The rms value of current in an ac circuit is 10 A. What is the peak current?

AC VOLTAGE APPLIED TO AN INDUCTOR

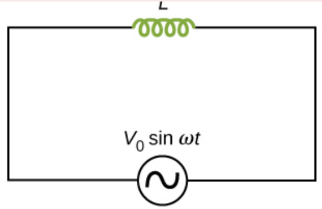


- Consider a pure inductor (having zero ohmic resistance) of inductance L connected to an alternating source.
- Let, the instantaneous value of alternating EMF be given by $\boxed{V = V_0 \sin \omega t}$ ....(1).
- Now, instantaneous voltage across inductor is equal to $\left(-L\frac{dI}{dt}\right)$. Using Lenz's law, we get

$$V - L\frac{dI}{dt} = 0 \text{ Or } V = L\frac{dI}{dt}$$

AC VOLTAGE APPLIED TO AN INDUCTOR

$$dI = \frac{V}{L}dt = \frac{V_0}{L}\sin\omega t\, dt$$



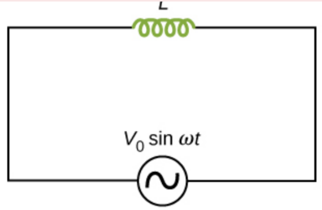
- Integrating we get

$$\int dI = \frac{V_0}{L}\int \sin\omega t\, dt$$

$$I = \frac{V_0}{\omega L}(-\cos\omega t)$$

$$I = -\frac{V_0}{\omega L}\sin\left(\frac{\pi}{2} - \omega t\right) \qquad \because \cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$$

AC VOLTAGE APPLIED TO AN INDUCTOR



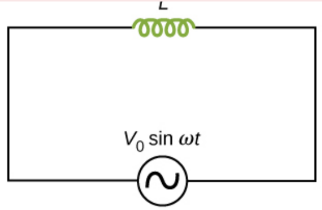
$$\because -\sin\theta = \sin(-\theta)$$

$$\boxed{I = \frac{V_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right) = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)} \quad ....(2)$$

- where $I_0 = \frac{V_0}{\omega L}$. The quantity ωL is **analogous to the resistance** and is **called inductive reactance**, denoted by X_L:

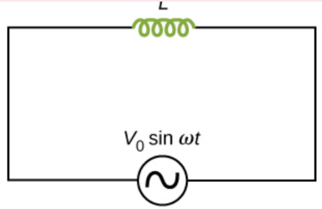
$$\boxed{\frac{V_0}{I_0} = \omega L = X_L} \quad .....(3)$$

AC VOLTAGE APPLIED TO AN INDUCTOR

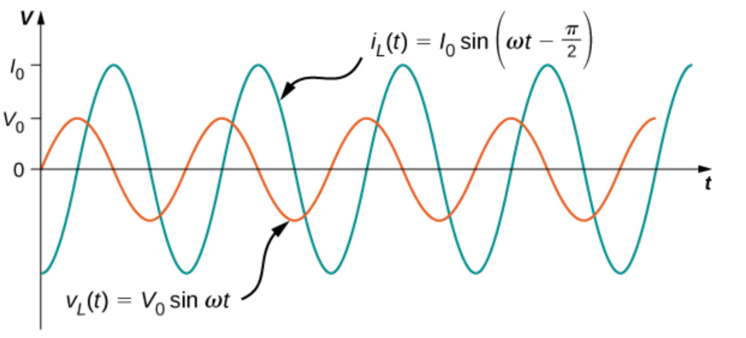


- The **dimension of inductive reactance is the same as** that of **resistance** and **its SI unit is ohm Ω**.
- The **inductive reactance limits the current in a purely inductive circuit** in the **same way as the resistance limits the current in a purely resistive circuit**.
- The **inductive reactance is directly proportional to** the **inductance** and to the **frequency** of the current.

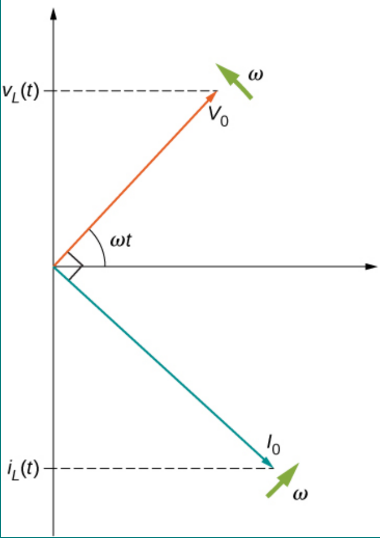
AC VOLTAGE APPLIED TO AN INDUCTOR



- Comparing equation (1) and (2) for the source voltage $\boldsymbol{V}$ and the current $\boldsymbol{I}$ in an inductor shows that the **current lags the voltage by** $\boldsymbol{\frac{\pi}{2}}$ or **one-quarter (1/4) cycle**.
- The **current phasor I** is $\boldsymbol{\frac{\pi}{2}}$ **behind the voltage phasor V**.



V
I_0
V_0
0
t
$i_L(t) = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$
$v_L(t) = V_0 \sin \omega t$



$v_L(t)$
ω
V_0
ωt
I_0
$i_L(t)$
ω

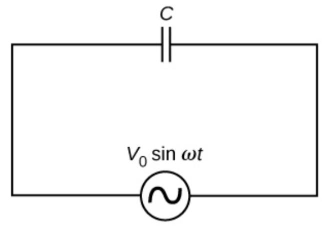
Example 2

A pure inductor of $\mathbf{25.0}\ \boldsymbol{mH}$ is connected to a source of $\mathbf{220}\ \boldsymbol{V}$. Find the inductive reactance and rms current in the circuit if the frequency of the source is **50 Hz**.

Exercise 3

A $\boldsymbol{44\ mH}$ inductor is connected to $\boldsymbol{220\ V,\ 50\ Hz}$ ac supply. Determine the rms value of the current in the circuit.

AC VOLTAGE APPLIED TO A CAPACITOR



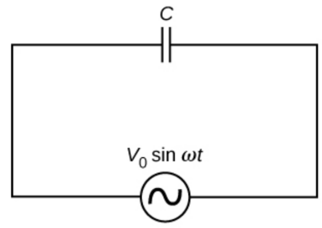
- Consider a **capacitor of capacitance *C*** connected to an alternating source.
- Let ***q* be the charge** on the capacitor **at any time t**. The **instantaneous voltage *V*** across the capacitor is

$$V = \frac{q}{C} \quad ....(1)$$

- From the Kirchhoff's loop rule, the voltage across the source and the capacitor are equal,

$$V_0 \sin \omega t = \frac{q}{C} \text{ Or } q = CV_0 \sin \omega t$$

AC VOLTAGE APPLIED TO A CAPACITOR



- Differentiating above equation w.r.t. t, we have v

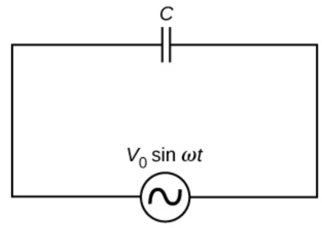
$$I = \frac{dq}{dt} = CV_0 \frac{d \sin \omega t}{dt} = \omega CV_0 \cos \omega t$$

- Using the relation, $\cos \omega t = \sin\left(\omega t + \frac{\pi}{2}\right)$, we have

$$I = \omega CV_0 \sin\left(\omega t + \frac{\pi}{2}\right) \text{ or}$$

$$\boxed{I = I_0 \sin\left(\omega t + \frac{\pi}{2}\right)} \ldots(2)$$

AC VOLTAGE APPLIED TO A CAPACITOR

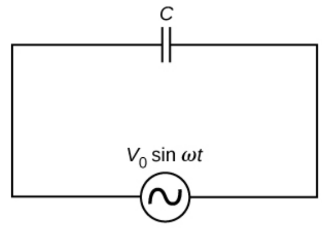


- where the amplitude of the oscillating current is $I_0 = \omega C V_0$. We can rewrite it as $V_0 = I_0 \frac{1}{\omega C}$
- Comparing it to $V_0 = I_0\ R$ for a purely resistive circuit, the quantity $\frac{1}{\omega C}$ is **analogous to the resistance** and is **called capacitive reactance**, denoted by X_C:

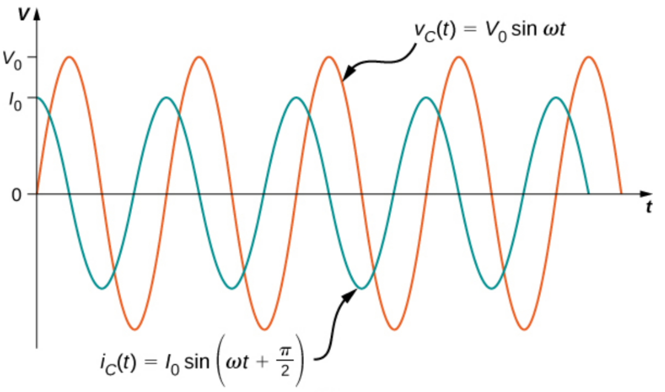
$$\frac{V_0}{I_0} = X_C = \frac{1}{\omega C}$$

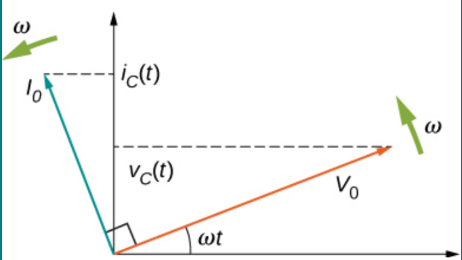
- **capacitive reactance is inversely proportional to** the **frequency and the capacitance**.

AC VOLTAGE APPLIED TO A CAPACITOR



- Comparing equation (1) and (2) for the source voltage V and the current I in a capacitor shows that the **current is ahead the voltage by** $\frac{\pi}{2}$ or **one-quarter (1/4) cycle**.
- The **current phasor I** is $\frac{\pi}{2}$ **ahead the voltage phasor V**.





https://www.animations.physics.unsw.edu.au/jw/AC.html

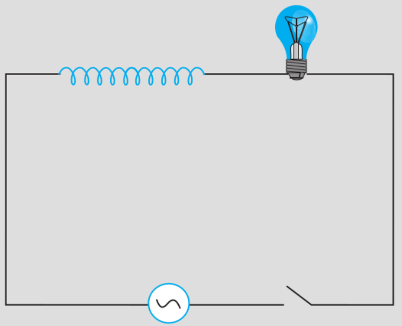
Example 3

A lamp is connected in series with a capacitor. Predict your observations for dc and ac connections. What happens in each case if the capacitance of the capacitor is reduced?

Example 4

A $15.0\ \mu F$ capacitor is connected to a **220 V, 50 Hz** source. Find the capacitive reactance and the current (rms and peak) in the circuit. If the frequency is doubled, what happens to the capacitive reactance and the current?

Example 4



A light bulb and an open coil inductor are connected to an ac source through a key as shown in Fig. The switch is closed and after sometime, an iron rod is inserted into the interior of the inductor. The glow of the light bulb (a) increases; (b) decreases; (c) is unchanged, as the iron rod is inserted. Give your answer with reasons

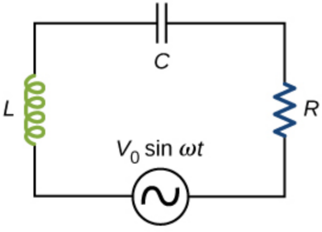
Exercise 4

A **60** $\boldsymbol{\mu F}$ capacitor is connected to a **110 V, 60 Hz** ac supply. Determine the rms value of the current in the circuit.

Exercise 5

In Exercises 3 and 4, what is the net power absorbed by each circuit over a complete cycle. Explain your answer

AC VOLTAGE APPLIED TO A SERIES LCR CIRCUIT

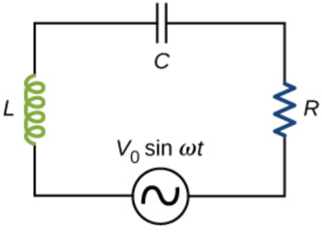


- Figure shows a **series LCR circuit connected** to an **ac source**. Let, the **voltage of the source** to be $V = V_0 \sin \omega t$
- If q is the **charge on the capacitor** and I the **current, at time t**, we have, from **Kirchhoff's loop rule**:

$$V = L\frac{dI}{dt} + \frac{q}{C} + IR \; \ldots.(1)$$

- **We want to determine the instantaneous current** I **and its phase relationship to the applied alternating voltage** V. **using** the **technique of phasors.**

Phasor-diagram solution

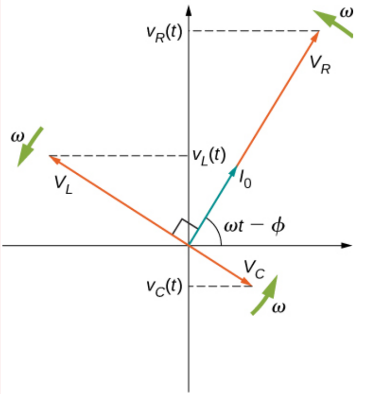


- Since the elements(L, C R) are in **series**, the **same current flows through each element** at all points in time. Let it be

$$\boldsymbol{I = I_0 \sin(\omega t - \phi) \quad .....(2)}$$

- where $\boldsymbol{I_0}$ is the **current amplitude** and ϕ is the **phase angle between the current and the applied voltage**.
- Our **task** is to **find** $\boldsymbol{I_0}$ and ϕ.
- A **pharos diagram involving** $\boldsymbol{I(t)}$, $\boldsymbol{V_R(t)}$, $\boldsymbol{V_C(t)}$, and $\boldsymbol{V_L(t)}$ is **helpful for analyzing** the **circuit**. As shown in Figure

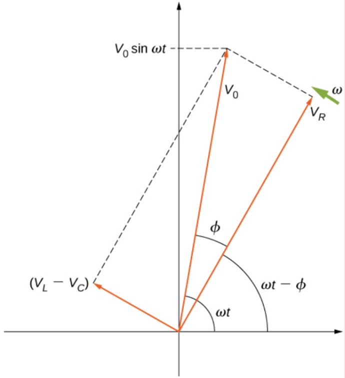
Phasor-diagram solution



The phasor diagram for the *RLC* series circuit

- The phasor representing $\boldsymbol{V_R(t)}$ points in the same direction as the phasor for $\boldsymbol{I(t)}$; its amplitude is $\boldsymbol{V_R = I_0 R}$ ...(3).
- The $\boldsymbol{V_C(t)}$ phasor lags the $\boldsymbol{I(t)}$ phasor by $\boldsymbol{\frac{\pi}{2}}$ rad and has the amplitude $\boldsymbol{V_C = I_0 X_C}$ ...(4).
- The phasor for $\boldsymbol{V_L(t)}$ leads the i(t) phasor by $\boldsymbol{\frac{\pi}{2}}$ rad and has the amplitude $\boldsymbol{V_L = I_0 X_L}$ ...(5).

Phasor-diagram solution

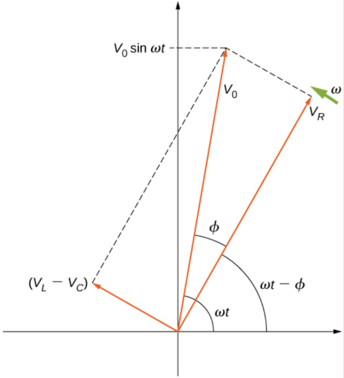


The resultant of the phasors for $v_L(t)$, $v_C(t)$, and $v_R(t)$ is equal to the phasor for $v(t) = V_0 \sin \omega t$.

- Since V_C and V_L are **always along the same line and in opposite directions**, they can be **combined into a single phasor** $(\vec{V}_C+\vec{V}_L)$ which has a magnitude $|V_L - V_C|$.
- The **vector sum of the phasors** is shown in Figure. The **resultant phasor has an amplitude** V_0 and is **directed at an angle** ϕ with respect to the $V_R(t)$, ***or*** $i(t)$, phasor.

$$\vec{V}_0=\vec{V}_R+\vec{V}_L+\vec{V}_C$$

Phasor-diagram solution



The resultant of the phasors for $v_L(t)$, $v_C(t)$, and $v_R(t)$ is equal to the phasor for $v(t) = V_0 \sin \omega t$.

- Since V_0 **is** represented as the **hypotenuse** of a right-triangle whose **sides** are V_R and $(V_L - V_C)$, the Pythagorean theorem gives:

$$\boldsymbol{V_0^2 = V_R^2 + (V_L - V_C)^2}$$

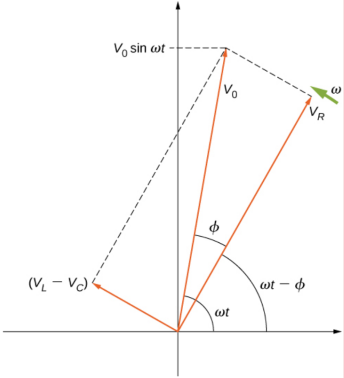
- Substituting the values of V_R, V_L and V_C form eq (3-4)

$$\boldsymbol{V_0^2 = (I_0 R)^2 + (I_0 X_L - I_0 X_C)^2}$$

$$\boldsymbol{V_0^2 = (I_0)^2 \left[R^2 + (X_L - X_C)^2\right]}$$

Or, $$\boxed{\boldsymbol{V_0 = I_0 \sqrt{R^2 + (X_L - X_C)^2}}} \quad \boldsymbol{\ldots.(6)}$$

Phasor-diagram solution



The resultant of the phasors for $v_L(t)$, $v_C(t)$, and $v_R(t)$ is equal to the phasor for $v(t) = V_0 \sin \omega t$.

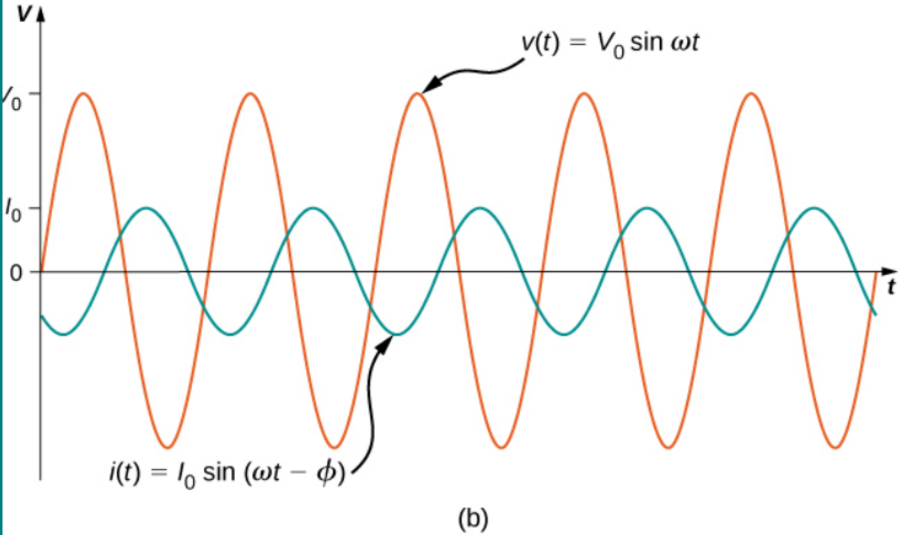
Or, $\boxed{V_0 = I_0 Z}$ ....(7)

- By **analogy to the resistance** in a circuit $(V = IR)$, we introduce the **impedance Z** in an ac circuit:

$$\boxed{Z = \sqrt{R^2 + (X_L - X_C)^2}} \quad ....(8)$$

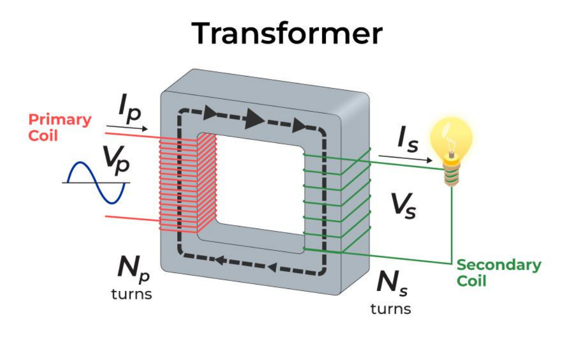
- For the phase angle ϕ

$$\boxed{\tan \phi = \frac{V_L - V_C}{V_R} = \frac{I_0 X_L - I_0 X_C}{I_0 R} = \frac{X_L - X_C}{R}} \quad .(9)$$



A comparison of the generator output voltage and the current.

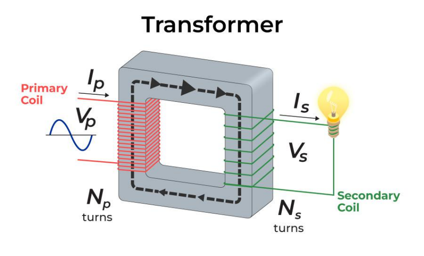
Transformers



- The **device that transforms** (increase or decreases) **A.C. voltages** from one value to another **using induction** is the transformer.
- Transformers is **based** upon the **principle of mutual induction**.

Transformers

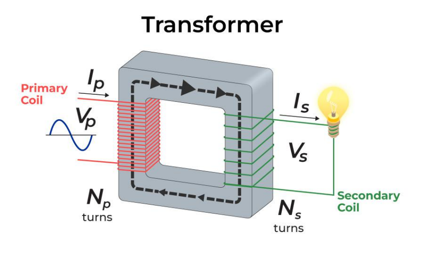
Construction:



- It **consists** of **two coils**, **primary coil** (p) and **secondary coil** (s), insulated from each other and **wounded on soft iron core**.
- The **primary coil has N_P** loops, or **turns**, and **is connected to an alternating voltage $V_P(t)$**.
- The **secondary coil has N_S turns** and is **connected to a load resistor R_S**.
- These **soft iron cores** are **laminated to minimise eddy current loss**.

Transformers

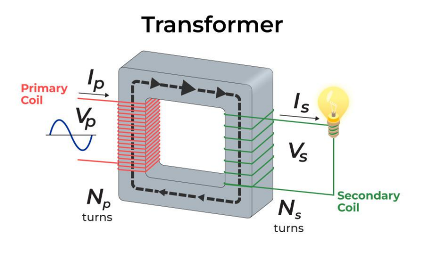
Assumptions:



- We assume the ideal case for which **all magnetic field lines are confined to the core** so that the **same magnetic flux permeates each turn of both the primary and the secondary windings**.
- We also **neglect energy losses to magnetic hysteresis**, to **ohmic heating in the windings**, and to **ohmic heating of the induced eddy currents** in the core.

Transformers

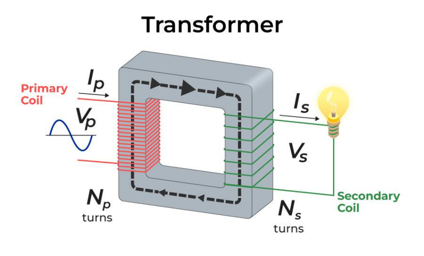
Working and Theory:



- When an **alternating voltage is applied to the primary**, the resulting **current produces an alternating magnetic flux which links the secondary coil** and **induces an emf in it**.
- The **input voltage $V_P(t)$** is **equal to** the **potential difference induced** across the **primary** winding. From Faraday's law, the **induced potential difference** is $-N_P\dfrac{d\phi}{dt}$

Transformers

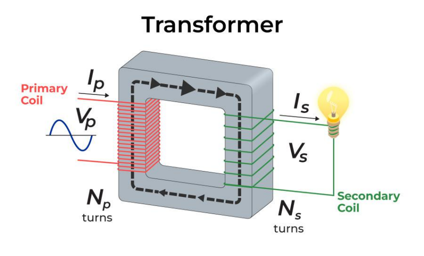
Working and Theory:



- where ϕ is the **flux through one turn of the primary winding**. Thus, $\boxed{V_P = -N_P \frac{d\phi}{dt}}$ ...(1)
- Similarly, the **output voltage $V_S(t)$** delivered to the load resistor must **equal the potential difference induced across the secondary winding**. Since the **transformer is ideal**, the **flux through every turn of the secondary winding is also ϕ**, and

Transformers

Working and Theory:



- $$\boxed{V_S = -N_S \frac{d\phi}{dt}} \ ...(2)$$
- Combining equation (1) and (2)

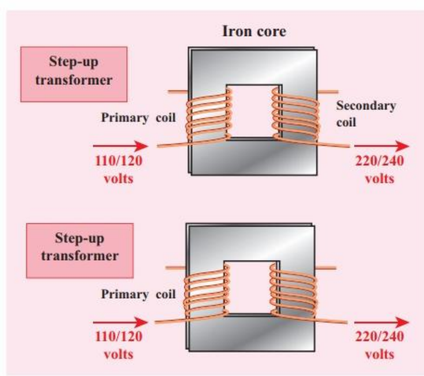
$$\boxed{\frac{V_s}{V_P} = \frac{N_S}{N_P}} \ ....(3)$$

- If the **transformer is assumed to be 100 %** efficient (no energy losses), the **power input is equal to the power output**

$$\boxed{I_S V_S = I_P V_P} \ ....(4)$$

Transformers

Working and Theory:



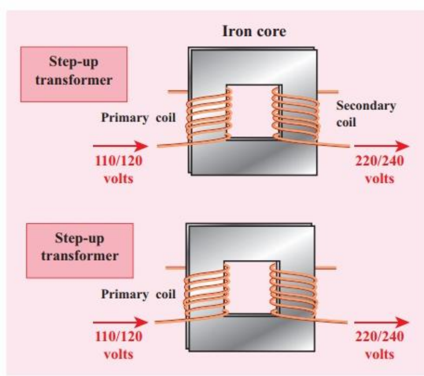
- Combining equation (3) and (4)

$$\boxed{\frac{V_S}{V_P} = \frac{I_P}{I_S} = \frac{N_S}{N_P}} \; ....(5)$$

- For a **step-up transformer, which increases voltage** $(V_S > V_P)$ and **decreases current** $(I_S < I_P)$, **this ratio** $\frac{N_S}{N_P}$ is **greater than one.**

Transformers

Working and Theory:



- For a **step-down transformer, which decreases voltage** $(V_S < V_P)$ and **increases current** $(I_S > I_P)$, **this ratio** $\frac{N_S}{N_P}$ is **less than one.**

Transformers

Energy Loss in Transformers:

In actual transformers, small energy losses do occur due to the following reasons:

- **Flux Leakage:** There is always some flux leakage; that is, **not all of the flux due to primary passes through the secondary** due to poor design of the core or the air gaps in the core. **It can be reduced by winding the primary and secondary coils one over the other**.
- **Resistance of the windings:** The wire used for the **windings has some resistance** and so, **energy is lost due to heat** produced in the wire I^2R. **In high current, low voltage windings, these are minimised by using thick wire**.

Transformers

Energy Loss in Transformers:

- **Eddy currents:** The **alternating magnetic flux induces eddy currents in** the **iron core and causes heating**. The effect is **reduced by using a laminated core**.
- **Hysteresis:** The **magnetisation of the core is repeatedly reversed** by the alternating magnetic field. The **resulting expenditure of energy in the core appears as heat** and is kept to a **minimum by using a magnetic material which has a low hysteresis loss**.